

Pixels to Predictions: Improving SmolVLM for Scientific Multiple-Choice Vision Reasoning

Anonymous ACL submission

Abstract

This project studies scientific multiple-choice vision reasoning under strict course constraints: no external data, offline evaluation, free-tier hardware, and fewer than five million trainable parameters. Starting from the official HuggingFaceTB/SmolVLM-500M-Instruct checkpoint, I first built a zero-shot baseline that uses the provided image, question, lecture, hint, and answer choices to predict a single answer index. I then evaluated several controlled inference modifications and lightweight fine-tuning strategies. The strongest approach was Low-Rank Adaptation (LoRA) with best-checkpoint selection and longer training, which improved validation accuracy from 0.6498 for the zero-shot baseline to 0.8025. The report emphasizes controlled ablations, reproducibility, and analysis of why some design choices helped while others did not.

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2 Introduction

Code: https://github.com/nicho1210/pixels_to_prediction

This final project addresses the *Pixels to Predictions: DL Vision Challenge*, a scientific multiple-choice reasoning task in which a vision-language model must read an image together with question context and output a single correct answer choice. The final submission format requires a CSV file with two columns, `id` and `answer`, where `answer` is a 0-indexed integer choice.

The task is challenging for two reasons. First, the model must combine visual information with long textual context, including a question, answer

choices, and optional supporting fields such as lecture and hint. Second, the system is evaluated by exact classification accuracy rather than free-form explanation quality, so even small formatting or decoding mistakes directly lower the score.

The course constraints strongly shaped the design. Only the provided competition data could be used. Training and inference had to remain feasible on free-tier resources, and the number of trainable parameters had to stay below five million. Because of these constraints, the project focused on a compact official base model, careful prompt design, output parsing, and parameter-efficient fine-tuning instead of full-model retraining.

3 Dataset

The competition data is provided as three CSV files and a corresponding image directory:

- `train.csv`: 3109 examples
- `val.csv`: 1048 examples
- `test.csv`: 1008 examples

Each example contains an image path, a multiple-choice science question, a JSON-encoded list of answer choices, and metadata such as hint, lecture, grade, subject, topic, category, and skill. Training and validation rows also include the ground-truth answer index.

3.1 Preprocessing

The `choices` field is stored as a JSON string in the CSV files, so it is parsed into a Python list before use. Images are loaded from disk, converted to RGB, and resized to 224×224 .

The baseline and LoRA systems use the same underlying data representation:

- `image`
- `question text`

075	• answer choices	121
076	• optional context fields: lecture and hint	122
077	For robustness, the inference pipeline includes a	
078	missing-image fallback that replaces an unavailable	
079	image with a blank white image of the same reso-	
080	lution. This was needed because two test images	
081	were missing from the local working copy. The	
082	fallback ensured that the pipeline still produced a	
083	complete submission file instead of failing at infer-	
084	ence time.	
085	3.2 Prompt Formatting	
086	The prompt is formatted as a multimodal input be-	
087	ginning with the required <image> token, followed	
088	by instructions, context, the question, and the an-	
089	swer choices. The final answer is requested in the	
090	form of a single capital letter. The final baseline	
091	prompt is conceptually:	
092	<code><image></code>	
093	<code>You are solving a multiple-choice</code>	
094	<code>science question.</code>	
095	<code>Choose the single best answer.</code>	
096	<code>Reply with only one capital</code>	
097	<code>letter.</code>	
	optional context from lecture and hint	
098		
099	<code>Question: ...</code>	
100	<code>Choices:</code>	
101	<code>A. ...</code>	
102	<code>B. ...</code>	
103	<code>...</code>	
104	<code>Answer:</code>	
105	4 Method	
106	4.1 Base Model	
107	The base model is	
108	HuggingFaceTB/SmolVLM-500M-Instruct,	
109	loaded with AutoProcessor and	
110	AutoModelForVision2Seq. The system per-	
111	forms image-text-to-text inference, so the	
112	multiple-choice problem is cast as constrained	
113	answer generation.	
114	4.2 Zero-Shot Baseline	
115	The first complete baseline used the full prompt	
116	with lecture, hint, question, and choices, then gen-	
117	erated a short answer string and parsed the first	
118	valid answer letter into a 0-indexed class predic-	
119	tion.	
120	Two details were important for the baseline:	
	• Only newly generated tokens were decoded,	121
	rather than the full input-plus-output sequence.	122
	• The parser explicitly handled outputs such as A,	123
	A. the, A) ..., and Answer: A.	124
	The baseline validation accuracy on the full vali-	125
	dation set was 0.6498.	126
	4.3 LoRA Fine-Tuning	127
	After the zero-shot baseline, I used parameter-	128
	efficient fine-tuning to adapt the model to the task	129
	while staying within the trainable-parameter bud-	130
	get. LoRA was attached to the attention projection	131
	layers:	132
	• q_proj	133
	• k_proj	134
	• v_proj	135
	• o_proj	136
	The LoRA configuration was:	137
	• rank $r = 8$	138
	• LoRA alpha = 16	139
	• dropout = 0.05	140
	• bias = none	141
	The training objective masked the prompt tokens	142
	and computed loss only on the answer portion of	143
	each sample. In other words, the model was trained	144
	to predict the correct answer letter given the image	145
	and the full prompt context.	146
	4.4 Training Setup	147
	The final best-performing training setup used:	148
	• optimizer: AdamW	149
	• learning rate: 5×10^{-5}	150
	• batch size: 2	151
	• epochs trained: 4	152
	• evaluation: full validation set after each epoch	153
	• model selection: best validation checkpoint	154
	During inference, the generation length was re-	155
	duced to a single token (max_new_tokens=1) to	156
	force the model to output only the answer letter	157
	instead of longer explanations.	158

Component	Setting	5.1 Prompt Ablations	189								
Base model	SmolVLM-500M-Instruct	Prompt ablations were evaluated on the same 200-example validation subset using the same model and decoding configuration.	190								
Image size	224		191								
Prompt answer format	single capital letter		192								
LoRA rank	8										
LoRA alpha	16	<table><tr><th>Prompt variant</th><th>Accuracy on</th></tr><tr><td>Full context (lecture + hint + question + choices)</td><td>0.3000</td></tr><tr><td>Hint only</td><td>0.2000</td></tr><tr><td>Minimal (question + choices only)</td><td>0.1000</td></tr></table>	Prompt variant	Accuracy on	Full context (lecture + hint + question + choices)	0.3000	Hint only	0.2000	Minimal (question + choices only)	0.1000	
Prompt variant	Accuracy on										
Full context (lecture + hint + question + choices)	0.3000										
Hint only	0.2000										
Minimal (question + choices only)	0.1000										
LoRA dropout	0.05										
LoRA target modules	q,k,v,o projections										
Optimizer	AdamW										
Learning rate	5×10^{-5}										
Batch size	2	Table 2: Controlled prompt ablations using the same subset and decoding setup.									
Best epoch selected	4										
Inference generation length	1 token										

Table 1: Final best-performing configuration.

4.5 Justification of Design Choices

This section explains the major design decisions and why each one was chosen.

Full-context prompt. I tested several prompt variants because the competition provides both lecture and hint fields, and it was not obvious whether they help or distract the model. The full-context prompt was retained because it consistently outperformed shorter alternatives on controlled validation subset comparisons.

Single-letter output. The task is scored as exact multiple-choice classification, not free-form explanation. Therefore, forcing the model to output only one letter is better aligned with the evaluation metric and reduces parser errors.

LoRA instead of full fine-tuning. The competition explicitly limits trainable parameters and assumes free-tier resources. LoRA is a natural fit because it improves task adaptation while remaining lightweight.

Best-checkpoint selection. Validation accuracy was measured after each epoch because final-epoch performance is not always optimal. This choice directly improved the rigor and reproducibility of model selection.

5 Ablations and Experimental Rigor

One weakness of the midterm report was insufficient ablation depth and weak control of comparisons, so this project explicitly evaluates alternatives under controlled settings.

These comparisons show that the extra context is useful in this task. Removing lecture and hint both hurt performance, and the fully minimal prompt degraded accuracy substantially.

5.2 Inference Strategy Ablation

I also tested constrained answer scoring, where each answer choice was scored rather than freely generated. On the exploratory validation subset, this method achieved 0.3000 accuracy on 20 examples, which did not improve over the generation-based setup used at the same stage. Therefore, it was not adopted as the final approach.

5.3 Training Ablation

The strongest gains came from task adaptation through LoRA and from training-schedule refinement:

System	Validation accuracy
Zero-shot baseline	0.6498
Initial LoRA model	0.7347
LoRA + best checkpoint	0.7653
LoRA + longer training + best checkpoint	0.8025

Table 3: Main model progression on the full validation set.

This progression supports two conclusions. First, task-specific adaptation matters more than the simple inference-only changes explored earlier. Second, longer training was beneficial in this setting because validation accuracy kept improving through epoch 4.

5.4 Fairness of Comparisons

To ensure fair comparisons:

- prompt ablations used the same 200-example validation subset,
- decoding settings were kept fixed across prompt variants,
- the same base model was used for all prompt-only experiments,
- LoRA comparisons used the full validation set,
- model selection was based on validation accuracy rather than subjective output inspection.

6 Results

6.1 Quantitative Results

The best validation result obtained so far is 0.8025 using LoRA fine-tuning with best-checkpoint selection after four epochs.

Epoch	Average train loss	Validation accuracy
1	0.2478	0.7576
2	0.1950	0.7824
3	0.1596	0.7824
4	0.1392	0.8025

Table 4: Training trajectory for the best LoRA run.

The zero-shot baseline already produced a valid submission pipeline, but LoRA substantially improved task-specific answer selection. The increase from 0.6498 to 0.8025 suggests that the model benefits from supervised adaptation to the competition distribution.

6.2 Runtime

Zero-shot validation and test inference were feasible but slow because predictions were generated example by example. LoRA training remained manageable on free-tier hardware because only a small number of parameters were updated and the batch size was kept small.

6.3 Leaderboard

At the time of this draft, the report records the best local validation accuracy and the complete submission-generation pipeline. The public leaderboard score can be inserted here once the final selected submission is uploaded.

7 Discussion

The experiments reveal several useful lessons.

First, prompt design matters, but not every simplification helps. In this task, the lecture and hint fields carried useful information, and removing them reduced accuracy. This is an important result because it shows that the best prompt is not necessarily the shortest prompt.

Second, task adaptation had a larger effect than the earlier inference-only modifications. The biggest gains came from LoRA fine-tuning and from selecting the best checkpoint based on validation accuracy.

Third, controlled experiments are essential. Some ideas that sounded promising, such as choice scoring or aggressive prompt simplification, did not improve performance when tested fairly.

7.1 Error Analysis and Limitations

Even the best model remains imperfect. Typical failure modes include:

- predicting a plausible but wrong answer letter when multiple choices are semantically similar,
- over-relying on textual priors when the image is subtle or ambiguous,
- sensitivity to local dataset issues such as missing test images in the working copy.

There are also methodological limitations. The prompt ablations used a 200-example subset for efficiency, so although they were controlled, they are smaller-scale than the full validation runs. In addition, the current report emphasizes local validation accuracy; final competition ranking depends on hidden test labels and may not perfectly match local trends.

8 Conclusion

This project developed a scientific multiple-choice vision reasoning system under strict resource and data constraints. Starting from the official SmoIVLM-500M-Instruct checkpoint, I first built a full zero-shot inference pipeline that produced valid competition submissions. I then improved the system using controlled prompt experiments and LoRA fine-tuning.

The most important findings are as follows. First, the full-context prompt worked better than shorter prompt variants, so the lecture and hint

fields should not be removed casually. Second, inference-only modifications yielded limited improvements. Third, LoRA fine-tuning produced the largest gains, increasing validation accuracy from 0.6498 to 0.8025 after best-checkpoint selection.

Overall, the project shows that a compact vision-language model can be adapted effectively to a scientific multiple-choice reasoning task without exceeding the course constraints. Future work can explore broader hyperparameter searches, stronger checkpoint-selection policies, and more detailed per-category error analysis to further improve leaderboard performance.

9 References

- Competition code repository: https://github.com/nicho1210/pixels_to_prediction
- Model: <https://huggingface.co/HuggingFaceTB/SmolVLM-500M-Instruct>